Purdue University
Davidson School of Chemical Engineering
Process Dynamics and Control

CHE 456 - Fall 2025

Instructors: **Professor David E. Bernal Neira**
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Email: dbernaln@purdue.edu
Office Hours: Will be organized online or in person based on requests and by appointment

Teaching Assistants:		**TA Office Hours:**	**Office locations:**
Shriram Suyog Joshi	joshi267@purdue.edu	M 9:00 – 11:00	FNRY B142A
Anurag Ramesh	rames102@purdue.edu	W 13:00 – 15:00	FRNY B142A
Zachary Hillman	zhillma@purdue.edu	T/Th 10:15 – 11:15	FRNY B142A
Akshdeep Singh Ahluwalia	ahluwala@purdue.edu	F 11:30 – 13:30	FRNY B142A

Please start all email communication subject line with "CHE456" (Capital letter, No space).
If you do not start the subject line with this, your email may get missed and not get answered.

Schedule:

- **Lectures:** Regular in-person class

  4:30 pm - 5:45 pm, Tuesdays and Thursdays MATH 175 Aug 26th-Dec 11th, 2025

- Lectures will generally be in-person, with a few online or recorded lectures during the semester. For the schedule of topics, see the table at the end of the syllabus.

- **PSO:** Practice study observation sessions are organized based on groups. These are scheduled as follows:
  M 9:30 am - 11:20 am (Anurag Ramesh)
  M 11:30 am - 1:20 pm (Shriram Suyog Joshi)
  T 11:30 am - 1:20 pm (Zachary Hillman)
  T 1:30 pm - 3:20 pm (Akshdeep Singh Ahluwalia)

All these sessions are held in Forney Hall, Room 1022.
PSO sessions are optional, but completing the PSO assignments earns extra credit. During the time allotted for the PSO, a TA will be available to provide support with the PSO tutorials and may occasionally use the time for office hours/problem classes; hence, attendance is recommended.

The PSO assignment will be due on Friday at 6 pm of the week when the session occurs. For the PSO session, you will need Python, either on your computer or via Google Colab. Please ensure these are installed and available. You can also use any remote access supported by the University to do the PSO assignments if you don't have Python installed.

Office hours:
You have multiple office hours and recitation sessions scheduled during the week/with recitation offered by the TAs, either remotely, face-to-face, or a combination of both. We will also hold office hours both online and in person. We will also organize these throughout the semester based on questions we receive via email or by request.

Grading Policy:

Homework Problems	25%
Midterm Exams 1	25%
Midterm Exams 2	25%
Final Exam	25%

Extra credit opportunities:

- You will receive up to an **extra 10%** for the PSO sessions, which will be added to the final average grade before final letter grades are decided.

NO MAKE-UP EXAMS.

Textbook: (Recommended)
Process Dynamics and Control (3rd Edition)
Dale E. Seborg, Thomas F. Edgar, Duncan A. Mellichamp, and Frank Doyle

Additional Course Materials:
A set of PowerPoint slides and other handouts covering the lecture materials will be made available online. These will be useful for preparing for the exams.

A *vade mecum*, including the most important equations and formulas, will also be provided.

Computational Resources
Python is available in the CHE computer lab. Handouts will be provided to introduce you to Python and related packages.

Course Objectives:
Develop a fundamental understanding of process dynamics and control using logical problem-solving techniques and computer laboratory exercises, and apply these concepts to chemical engineering (or related) situations.

Course Outcomes:
1. Identify a process, understand the problem's physics, develop a mathematical model as a surrogate for the process, and solve the problem analytically or numerically using computer packages.
2. Understand the building blocks of process control and synthesize them to provide complete process control solutions.
3. Design, model, and test a hierarchical control system on a complex process.
4. Understand the role of process control in meeting stringent safety, economic, environmental, societal, and corporate goals.
5. Work effectively to execute process control in computer laboratories professionally and ethically.

Course Topics:

Laplace transforms; solution of differential equations; process models (linear, nonlinear, deviation variable, frequency response, transform-domain); first order, second order, nth order transfer functions; inverse response systems; time-delay systems; Bode diagrams; frequency-response analysis; model formulation; step-response identification; block diagram representation; classical controllers; closed-loop transient response; proportional control offset; characteristic equation; Routh's stability test; stability by direct substitution; root locus diagram; strategic principles of control systems; performance criteria; controller tuning; cascade control; feed-forward control; feedback-augmented feed forward control; internal model control, smith predictor; basic concept of model predictive control; optimization.

Specific course information

a. **Catalog description:** Dynamic response and control of chemical processing equipment, such as heat exchangers, chemical reactors, and absorption towers. Fundamental techniques of servomechanism theory, including block diagrams, transfer functions, and frequency response, are utilized. Introduction to advanced control techniques.
b. **Prerequisites:** Undergraduate level CHE 37700 Minimum Grade of C- and Undergraduate level CHE 34800 Minimum Grade of C- [may be taken concurrently] and Undergraduate level CHE 37800 Minimum Grade of C- [may be taken concurrently]
c. **Course Status:** Required

Specific goals for the course

a. **Specific outcomes of instruction**
   1. Develop a physically relevant mathematical model as a surrogate for a chemical process (20500, 21100, 30600, 34800, 37700, 37800→); (→45000)
   2. Apply appropriate analytical and computer packages for analyzing the process control behavior based on the developed model (20500, 21100, 30600, 34800, 37700, 37800→); (→45000)
   3. Design, model, and test a hierarchical control system on a complex process (20500, 21100, 30600, 34800, 37700, 37800→); (→45000)
   4. Describe the role of process control in meeting stringent safety, economic, environmental, societal, and corporate goals (20500, 21100, 30000, 30600, 34800, 37700, 37800, 40000→); (→45000)
   5. Describe the building blocks of process control and their synthesis for providing complete process control solutions (→45000)
   6. Work effectively in a team to execute process control computer-based assignments in a professional and ethical manner (20500, 30600, 34800, 37700, 37800→); (→45000)

b. **Student outcomes addressed by the course:**

1) Apply principles of engineering, science, and mathematics to solve complex chemical engineering problems.
2) Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
5) Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
7) Acquire and apply new knowledge as needed, using appropriate learning strategies.

Course Structure:

The lectures will aim to define key theoretical concepts of process control; some examples will be discussed. Several lectures will be conducted as problem-solving classes, focusing on practical problems.

Homework Problems:

Homework problems will be assigned approximately every two weeks, depending on the material covered. The exact due date for each homework assignment will be provided. The problems are designed primarily to help you learn the course material and to prepare you for the exams.

Typically, there will be about 3-5 problems per homework assignment. To keep the grading load under control for such a large class, the grading of homework assignments will be as follows:

- All problems will be graded based on the correct final results. Please double-check your calculations and final results, as Homework problems will be graded based on CORRECT FINAL RESULTS ONLY.
- Additionally, one randomly selected problem will be examined in detail, and the solution process will be evaluated. The final correct answer will only be accepted if the correct steps are outlined and followed.

One of the last homework assignments will be to fill in the course evaluation forms. To receive credit for this homework, you must submit the confirmation page bearing your name, confirming that the evaluation has been submitted.

Homework will be released on Wednesday morning and is due on the following Tuesday at 10:00 pm. You will have to submit it online via Brightspace.

For every hour an assignment is late, 20% of the total points earned on this homework will be deducted. For example, if the total possible points are 10 and the assignment is submitted 1 hour and 5 minutes late, then 4 points will be deducted. If a student initially scored 7 points, they would end up with 3 points.

Midterm Exams:

Two one-and-a-half-hour midterm exams will be given during the scheduled time. This will be organized online. The exam will be released at a given time, and you will have a fixed time period to submit your answers. The exam period will overlap with the class time.

Final Exam:

This will be similar to the midterm exams, except it will cover **all the course material**.

Revised Grades:

Invariably, questions arise concerning the fairness of a grade on homework or exams. The following policy applies to:

i. The entire homework or exam is subject to revision in the grade, not just a single problem.
ii. **The discussion regarding a grade must be initiated within one week of the date the grade was released on Brightspace (for the final exam, on the same day).**

To initiate a re-grading, a written request must be submitted to the lead TA. If the re-grading applies to the final exam, then the request should be made to Professor Bernal.

Practice Study Observation/Recitation Sessions:

There will be PSO/recitation sessions organized during the allocated times. During these recitation sections, some supplementary class materials will be covered, enhancing your understanding of the key concepts discussed in the lecture. Additionally, an introduction to Python will be provided, accompanied by computer exercises to help you understand the course material. **These sessions are optional** but highly recommended. You will receive extra credit for taking part in these sessions:

- 50% will be based on participation and attendance.
- 50% will be based on the assignments given out for the PSO sessions.

CHE 456 Webpage:

Course material and announcements will be posted on **Brightspace**. Visit and log in with your Purdue career account/password. We will use this page to post important class announcements, as well as hints and solutions for homework assignments, recorded online lectures, and other relevant materials. It is certainly important to check this page regularly.

Absences:

No excuses for missed exams will be accepted, except for certified medical excuses or letters written on university letterhead for official school/university activities.

Job interview is not an acceptable excuse. PLEASE DO NOT REQUEST MAKE-UP EXAMS FOR THIS OR SIMILAR REASONS. The course assessment structure is designed in a way that you can obtain extra credits and get maximum grade in the course even if you must miss certain components of the assessment.

Academic Honesty:

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information submitted, the greater the opportunity for the University to investigate the concern. More details are available on our course Brightspace under University Policies and Statements.

Students are individually responsible for each homework, quiz, laboratory module, and exam. Any form of cheating will result in an F grade for the entire course for all parties involved. While discussions of homework and laboratory modules among classmates are to be expected, students are responsible for individually submitting their own work on all assignments. Thus, homework or laboratory modules that appear to have been copied are considered a form of cheating. Any form of collaboration on exams or quizzes is considered cheating.

Artificial Intelligence policy:

The prevalence of large language models (LLMs) is a reality that affects us all. For this reason, learning to use AI is an emerging skill that we all need to acquire. Please be aware of the limits of these tools and consider the following:

- If you provide minimum-effort prompts, you will get low-quality results. You will need to refine your prompts to achieve good outcomes. This will take effort.
- Don't trust anything it says. If it provides you with a number or fact, assume it is incorrect unless you either know the answer or can verify it with another source. You will be responsible for any errors or omissions provided by the tool. It works best for topics with which you are familiar.
- AI is a tool, but one that you need to acknowledge using. Please include a paragraph at the end of any assignment where you use AI, explaining what you used the AI for and what prompts you used to get the results (save the prompts, as they might be required for revision). Failure to do so violates academic honesty policies.

Be thoughtful about when this tool is useful. Don't use it if it isn't appropriate for the case or circumstance."

Mental Health and Wellness:

We strive to make this course and your time at Purdue University an enjoyable and nourishing learning experience. Therefore, we encourage you to prioritize your physical and mental well-being.

If you find yourself beginning to feel some stress, anxiety, and/or feeling slightly overwhelmed, try WellTrack**.** Sign in and access information and tools at your fingertips, available to you anytime.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are Monday through Friday, 8:00 am - 5:00 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign-up is free and can be done on BoilerConnect.

If you're struggling and need mental health services, Purdue University is committed to promoting the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or needing mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Any student who faces challenges securing food or housing and believes this may impact their performance in the course is encouraged to contact the Dean of Students for support. No appointment is necessary, and Student Support Services is available to serve students from 8:00 am to 5:00 pm, Monday through Friday.

Emergency Preparation:

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted on the course website or communicated to students by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email frequently.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website provides severe weather guidance, Emergency Plans, and a sign-up option for the Emergency Warning Notification System. We encourage you to download and review the *Emergency Preparedness for Classrooms document (**PDF**) or (**Word**).*

On the first day of class, we will review the **Emergency Preparedness plan for our specific classroom**, following Purdue's required Emergency Preparedness Briefing. Please make a note of items like:

- The location to which we will proceed after evacuating the building if we hear a fire alarm.
- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat, such as a shooting.

Purdue University is committed to maintaining a community that recognizes and values every person's inherent worth and dignity, fosters tolerance, sensitivity, understanding, and mutual respect among its members, and encourages everyone to strive for their full potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies and Statements.

Tentative Lecture Syllabus CHE456 (Fall 2025)

Week	Date	Topic	Recommended reading	Comments
1	8/26	Introduction to process control, feedback, and process dynamics	Ch 1	No PSO
1	8/28	Control Objectives (part 1)		
2	9/2	Control Objectives (part 2)		PSO – Python intro (1+2) (T and W sessions only, M sessions canceled due to Labor Day on Sep 1st)
2	9/4	Dynamic modeling (part 1)	Ch 2	HW1 released on W 9/3, due next T
3	9/9	**No class due to Industrial Roundtable**		PSO – Python Control Tutorial 1 M sessions only. T and W sessions canceled due to Industrial Roundtable.
3	9/11	Dynamic modeling (part 2)		HW2 released on W 9/10, due next T
4	9/16	Math Revision and Laplace Transforms	Ch 3	
4	9/18	Transfer functions – definition, properties	Ch 4	
5	9/23	Laplace Transform further properties	Ch 4	PSO – Python Control Tutorial 2
5	9/25	Transfer functions – deviation variables, linearization	Ch 4	HW3 released on W 9/17, due next T
6	9/30	First-order systems	Ch 5	PSO – Python Control Tutorial 3
6	10/2	Second-order systems	Ch 5	HW4 released on W 9/24, due next T
7	10/7	More complex systems	Ch 6	PSO – Python Control Tutorial 4
7	10/9	Problem class/Revision		
8	10/14	**Fall Break**		No PSO
8	**10/16**	**Exam 1**		
9	10/21	More complex systems		PSO – Python Control Tutorial 5
9	10/23	Frequency analysis of linear systems (part 1)	Ch 5, Ch 6, Ch 14.1-14.3	HW5 released on W 10/22, due next T
10	10/28	Frequency analysis of linear systems (part 2)		PSO – Python Control Tutorial 6
10	10/30	Block diagrams, closed-loop transfer function, and System stability	Ch 11	
11	11/4	PID control	Ch 8 (Ch 9)	PSO – Python Control Tutorial 7
11	11/6	PID control		HW6 released on W 11/5, due next T
12	11/11	PID controller tuning	Ch 12, Ch 14.6, Ch 14.7	PSO – Problem class
12	11/13	PID controller tuning		
13	11/18	Problem class/Revision		
13	**11/20**	**Exam 2**		
14	11/25	Cascade, feed-forward, and ratio control	Ch 15, Ch 16.1	No PSO
14	11/27	**Thanksgiving break**		
15	12/2	Time delay compensation and IMC-based PID tuning	Ch 16.2, Ch 12.3	PSO – Python Control Tutorial 8
15	12/4	Industrial guest lecture		HW7 released on W 12/3, due next T
16	12/9	Industrial guest lecture		No PSO
16	12/11	Revision		
	TBD	**Final Exam**		